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Section 1

Biology

Understanding of the effect of ancestral and natural saltwater treatment on durability, fibers densification and chemical modification of palm wood

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ABSTRACT

The palm-tree sector plays a very important role on both the socioeconomic and ecological levels, in Tunisia. There are three million trees in Tunisian palm plantations, ensuring a potential significant wood production, mainly in the craft and furniture industries.

In the past, Date Palm wood (*Phoenix dactylifera* L.) was also used as structural material. Its low natural durability and its low mechanical properties were improved by an ancestral preservative method consisting in the immersion of the trunk of the palm tree trunk freshly slaughtered for a period ranging from 1 to 2 years (depending on the species) in the salt waters of the Lake of Chot Djerid. This ancestral practice was disappeared, and it is always difficult to find more information on the different parameters involved in this kind of process.

The objective of this work was to assess the main technological qualities of palm wood preserved by salting while trying to retrace the steps of this natural and eco-friendly preservation process.

Two Common date palm cultivars woods (*Kentichi* and *Deglet Noor*) with ages ranging from 40- to 50-years, native and preserved by salting in the Chot Djérid were used for the experiments. Each wood samples were collected at the Regional Center of Research on Oasis Agriculture - Degache - Southern Tunisia. Densities (air-dried, water saturated, basic), mechanical properties, decay and termites resistances tests were performed on native and salt water - treated palm woods.

The first results showed a significant increase of the air-dried density of palm wood samples which increases after their wood salt water immersion. Salt water treatment allows improving greatly the palm wood MOR in bending and paralleling to the fibers. Termite's resistance tests highlighted that native and treated palm wood had a similar degradation level after termite exposure, but the termite mortality rate was higher for the treated wood than that of native palm wood. Extractives, lignin and cellulose contents are slightly more abundant in the control samples except for the hemicelluloses which are more abundant in the treated palm wood sample. Additional analyses conducted with an ICP-AES 700 Agilent device, evaluating the mineral compositions between native and salt-treated palm woods, give us some explanation ways about the performance improvement of wood after such a natural preservation process.

Keywords: palm wood, chemical modification, Brining treatment, decay and termite resistance, mechanical properties.

1. INTRODUCTION

Nowadays, the Tunisian primary wood-processing industry consumes more than 872,000 m³ of wood raw materials, of which only 119,000 m³ are supplied by local forest resources. In addition, several natural lignocellulosic fibers, as those from palm trees, are available around the country but they remain poorly exploited. The date palm (*Phoenix dactylifera* L.) is a tropical and subtropical tree that belongs to the palmae (*Arecaceae*) family and plays a very important role on both the socio-economic and ecological levels, mainly in southern Tunisia (Bendahou *et al.*, 2007). There are three million trees in Tunisian palm plantations, represented on an area of 22.500 ha (Rhouma, 1993), ensuring a potential significant wood production (Figure 1).

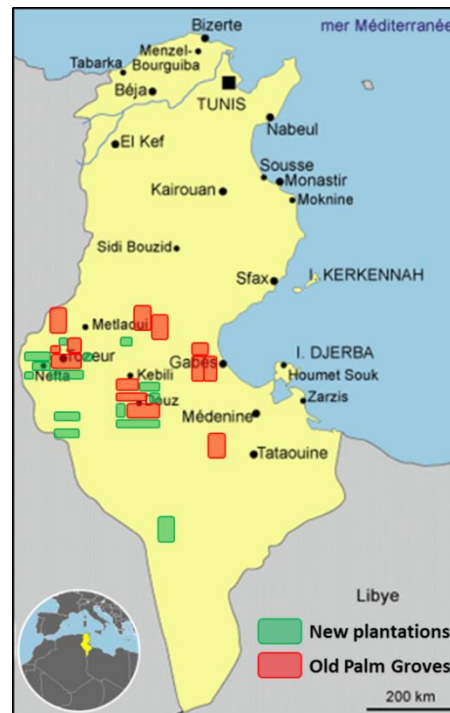


Figure 1: Repartition of new and old Palm Groves in Tunisia. ©Kévin Candelier

The Palms are also used in programs to control silting and they are extremely useful in controlling desertification by creating a microclimate which prevents long-term degradation of ecologically weak environments (Bendahou *et al.*, 2007).

Date palm (*Phoenix dactylifera* L.), one of the most important non-wood renewable resources in Maghreb area, used mainly for its fruit, fiber and as a construction material (Nasser, 2014), could also be considered a raw material for energy production (Nasser *et al.*, 2016), for the pyrolysis characteristics of the seeds (Babiker *et al.*, 2013), in briquette production (Eissa *et al.*, 2013), and for biochar production (Usman *et al.*, 2016). The residues are used in various industries as raw materials for many applications such as lumber (El-Mously, 1997), particleboard and oriented strand board (Hegazy and Ahmed, 2015), pulp and paper (Khiari *et al.*, 2011), wood-cement composites (Nasser and Al-Mefarrej, 2011) wood-plastic composites (Abu-Sharkh and Hamid, 2004).

For several centuries, palm stems have been used as a material for the construction and manufacturing throughout the Maghreb area. The stipe, also called palm tree convenience serves as a structure and carpentry in the construction of houses in the oasis in the southern Tunisia, in the manufacture of portals or to cover the roofs (Figure 2).



Figure 2: Date Palm stems, treated naturally by salt waters of the Lake of Chot Djerid, used as structural materials in the region of Djerid in Tunisia. ©Mohamed Elaieb

The ancients used fresh palm wood, initially with a very low natural durability and bad physical and mechanical quality. They treated it naturally by salt waters of the Lake of Chot Djerid (Figure 3). Such a treatment consists in depositing the trunk of the palm tree freshly slaughtered for a period ranging from 1 to 2 years (depending on the species), in the Chot.

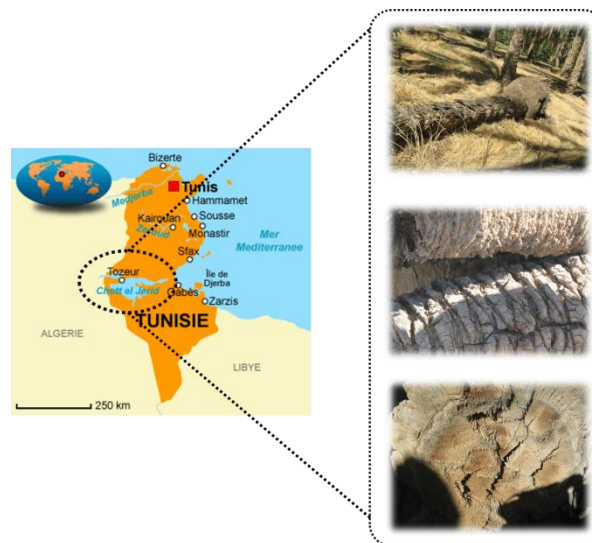


Figure 3: Exposition of date palm tree trunk, freshly slaughtered, into the salt waters cycles into the Lake of Chot Djerid. ©Kévin Candelier

After this stay, the material changes in appearance, colour, properties and the wood material becomes more dense, durable and resistant. Such a material could be used in outdoor construction, without any other industrial or expensive process. This ancestral practice was disappeared, and it is always difficult to find more information on the different parameters involved in this kind of process (durations, dimensions of the trunks, etc.). The improvement of living conditions has greatly influenced the population. Population replaced the palm wood construction with the concrete construction and thus abandoned the ancestral know-how. This method of treating this wood is not well known and there is little information that should be sought. Submerging wood in fresh water for prolonged periods can lead to extensive bacterial attack and marked increases in

permeability (Ellwood and Ecklund 1959). Repeated saltwater immersions alternating with drying steps can result in tracheid separation and bursting in the wood surface (Johnson et al. 1992), but the effects of this damage on durability are less clear. The effect of saltwater immersion on durability of wood has not been supported by substantial experimental data. It remains a lack of knowledge about wood properties characterisation after its long time salt water exposure, primarily because marine borers rapidly destroy such materials (Zabel and Morrell 1992). Those organisms are absent from the Lake Chot Djérid, making a study of the effects of prolonged salt exposure on durability possible. We tested date palm wood from the Southern Tunisia for resistance to fungal decay and for the distribution and composition of absorbed salts.

The objective of this work was to assess the main technological qualities of palm wood preserved by salting while trying to retrace the steps of this natural and eco-friendly preservation process. Two Common date palm cultivars woods (*Kentichi* and *Deglet Noor*) with ages ranging from 40- to 50-years, native and preserved by salting in the Chot Djérid were used for the experiments. Each wood samples were collected at the Regional Center of Research on Oasis Agriculture – Tozeur / Degache - Southern Tunisia (Figure 3). Densities (air-dried, water saturated, basic), mechanical properties, decay and termites resistance tests were performed on native and water salt treated palm woods. Additional analyses conducted with an ICP-OES 700 Agilent device, evaluating the mineral compositions between native and salt-treated palm woods, give us some explanation ways about the performance improvement of wood after such a natural preservation process.

2. MATERIALS AND METHODS

2.1 Palm wood samples selection

Two defect-free trees of common date palm cultivars (*Kentichi* and *Deglet Noor*) with ages ranging from 40 to 50-years and two samples (boards) preserved by salting in the Chot Djérid (*Kentichi* and *Deglet Noor*) taken from about 50 years old structure, were used for the experiments. The visual aspects of the selected palm wood samples are presented in Figure 4. All palms were harvested at the Regional Center of Research on Oasis Agriculture - Tozeur / Degache - Southern Tunisia (Figure 3). The site has the following characteristics: temperatures ranging between 10 °C in winter and 41 °C in summer, 50 mm annual rainfall and a calcareous soil. We can observe that after saltwater immersion, the both palm woods seem to be more dark, heavy, dense and homogeneous than the respective native palm wood samples.

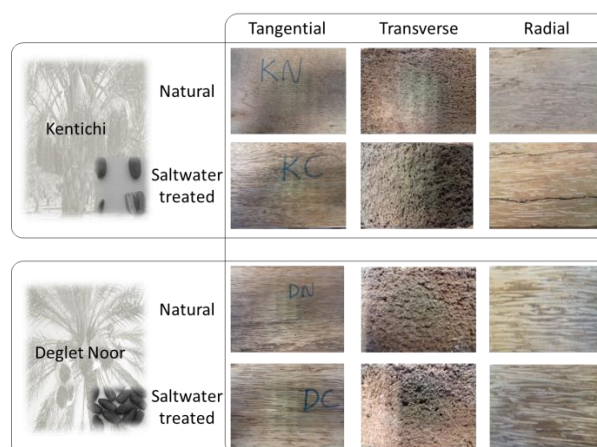


Figure 4: Appearance of natural and salt-treated palm wood samples, *Kentichi* and *Deglet Noor* varieties.

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2.2 Chemical composition

2.1.1 Extractives contents

Woods of each sample were ground and passed through different sieves to obtain particles ranging between 0.2 and 0.5 mm. This sawdust was successively washed in a Soxhlet extractor with a toluene/ethanol (2:1, v/v) mixture (6 h), followed by ethanol (6 h) and dried at 103 °C for 48 h.

2.1.2 Holocellulose contents

500 mg of extracted sawdust were placed in a 100 mL flask containing 30 mL of distilled water and heated at 75 °C. Acetic acid (0.1 mL) and 15% aqueous sodium chlorite (2 mL) were then added each hour for 7 h. The mixture was filtered on a Büchner funnel and the residue washed with water, soxhlet extracted for 2 h with ethanol and dried at 103 °C to a constant mass.

2.1.3 Hemicellulose and α -cellulose contents

The dried holocellulose prepared as above was placed into a 250 mL glass beaker covered with a glass cover containing 10 mL of 17.5% NaOH solution and maintained at 20 °C in a water bath. This mixture was agitated with caution with a rod glass until the holocellulose was soaked with the NaOH solution. 5 mL of 17.5% NaOH solution were then added every 5 min during 15 min and the mixture left for 30 min. The mixture was then diluted with 33 mL of distilled water and kept for 1 h before filtration on a Büchner funnel. The crude α -cellulose residue was successively washed with 100 mL of 8.3% NaOH solution, 100 mL of distilled water, 15 mL of 10% acetic acid solution and finally 250 mL distilled water. The residual α -cellulose was dried at 103 °C to a constant weight and the hemicellulose content calculated as follows:

$$\text{Hemicellulose (\%)} = \text{Holocellulose (\%)} - \text{Cellulose (\%)} \quad (1)$$

2.1.4 Klason lignin contents

500 mg of sawdust were mixed with 72% H₂SO₄ (10 mL) for 4 h at room temperature. The mixture was then diluted with 240 mL of distilled water, heated under reflux for 4 h and filtered. The residue of Klason lignin thus obtained was washed with hot water and dried at 103 °C to a constant weight.

2.3 Physical properties

To perform physical and mechanical tests:

- (i) Concerning control samples, a wooden disk 50 mm in thickness was cut at 1.40 m from the ground for each selected tree. To avoid errors during sampling, extreme cases such as excessively knotty trees or the presence of reaction wood or slope grain were taken discarded (ISO 4471, 1982). From each disk, 12 samples of 3 cm width from bark to bark were cut. These samples were then cut into strip 2 cm thick.
- (ii) For salt-treated date palm woods, similar samples were taken from previously collected boards.

Basic density (Db), Air-dry density (after conditioning in a climatic room at 20 ±2 °C and 65 ±5 % RH) and water-saturated (after drying in an oven at 103°C) density (D_{m12}, D_s) (ISO 13061-2, 2014), of the wood samples were determined using wood specimens of 20 × 20 × 10 mm (along the grain). The densities were determined by the gravimetric method (Haygreen and Bowyer, 1996).

$$D_b = \frac{m_0}{V_h} (2) \quad ; \quad D_s = \frac{m_h}{V_h} (3) \quad ; \quad D_{m12} = \frac{m_{12}}{V_{12}} \quad (4)$$

Where m_h is the humid mass of the initial sample and m_0 is the oven-dried mass of the wood sample. m_0 and m_{12} are the oven-dried and air-dried weight of the sample (g), respectively.

D_b is the basic density of wood (g.cm^{-3}), D_s is the water-saturated density of wood (g.cm^{-3}) and D_{m12} is the Air-dried density of wood (g.cm^{-3}). V_h is the green volume of the specimen (cm^3), V_0 is the oven-dried volume of the sample and V_{12} is the air-dried volume of wood sample.

2.4 Mechanical properties

To perform mechanical resistance tests, three point bending (MOR) and compression tests were carried out for each of the selected wood tree samples, and results were compared. An INSTRON 4467 Universal Mechanical Test Machine was used for the measurements. Samples were conditioned in a climate-controlled room with 65% RH and at 22 °C for the time required to stabilize the samples weights.

2.4.1 Bending test

Three point static bending tests were carried out according to the [EN 408 \(2010\)](#). The sample size was 300 x 20 mm x 20 mm³ (L x R x T). The moving head speed and span length were 0.09 mm.s⁻¹ and 260 mm, respectively. The load deformation data obtained were analyzed to determine the modulus of rupture (MOR). The tests were replicated on 20 samples from each selected Palm tree.

2.4.2 Compression strength parallel to grain

Compression tests were carried out according to the [EN 408 \(2010\)](#). Deviating from the norm, a reduced specimen size of 30 x 20 x 20 mm³ (L x R x T) was used. The moving head speed was 0.09 mm.s⁻¹ to ensure wood sample rupture within 1.5 to 2 minutes. The load deformation data obtained were analyzed to determine the modulus of rupture (MOR). 20 specimens per selected tree were tested.

2.5 Fungal resistance tests

Laboratory decay tests using specimens with surrounding dimensions of 30 x 10 x 5 mm³ (L x R x T) were obtained to examine the fungal resistance of the different date palm woods according to their salty water treatment. Beech (*Fagus sylvatica*) and pine sapwood (*Pinus sylvestris*) samples with the same dimensions have also been tested against fungi as controls. Sterile culture medium (65 ml), prepared from malt extract (40 g) and agar (20 g) in distilled water (1 L), was placed in a 72 cl volume glass bottle, closed with Carded cotton inoculated with two small piece of mycelium of a freshly grown culture of three brown-rots [*Coniophora puteana* (CP) (BAM ebw. 15); *Poria placenta* (PP) (FPRL 280) and *Gloeophyllum trabeum* (GT) (BAM Ebw. 109)]; and one white-rot [*Coriolus versicolor* (CV) (CTB 863 A)].

The inoculated Petri dishes were incubated for 2 weeks at 27°C and 70% RH until malt extract agar was covered by mycelium. Date palm woods and control specimens were supported on sterile inert mesh to prevent contact with culture medium. All test samples previously oven dried at 103°C to constant weight (m_0) were γ -ray-sterilized and three specimens and one control sample were placed in each test device under sterile conditions.

All modalities were triplicated and the incubation was carried out for 8 weeks at 22 °C under controlled humidity conditions of 70% RH in a climatic chamber. At the end of the test exposure, mycelia were removed and all samples were oven-dried to constant mass at 103°C (m_1). The extent of the fungal attack was determined based on the percentage of mass loss (ML) which was expressed as a percentage of the initial oven-dried weight of wood samples according to the following Eq. 5:

$$WL (\%) = \frac{m_0 - m_1}{m_0} \times 100 \quad (5)$$

2.6 Termite resistance tests

Palm wood samples were exposed to termites (*Reticulitermes flavipes*, ex. *santonensis*) in non-choice screening tests. Beech (*Fagus sylvatica*) and pine sapwood (*Pinus sylvestris*) samples of dimensions 30 x 10 x 5 mm³ (L x R x T) have also been tested against termite as controls. Prior to the test, each - sample and control samples (of the same dimensions) were dried at 103°C in order to obtain its anhydrous initial weight (m_2), and then re-conditioned at 20°C, 65% RH until constant weight. For each set of palm and control woods, three replicates were tested for their resistance towards termites. Each specimen was placed in the middle of a 9 cm diameter Petri dish containing 40 g of Fontainebleau wet sand (4 vol. of sand / 1 vol. of deionized water). A total of 50 termite workers, one nymph and one soldier were then introduced into each test device. These test devices were placed in a dark climatic chamber kept at 27°C and >75% RH for 4 weeks. Observations were carried out on a weekly basis in order to add water and check termite behavior (Figure 5).

At the end of the exposure, the samples were removed, cleaned of sand and termite survivors were counted. Sample degradations were given a visual rating according to the criteria of EN117 (2013) (the criteria being adjusted to the sample size). Then, the samples were dried at 103°C to obtain their anhydrous weight after termite exposure (m_3) and their weight losses (WL %) were calculated according to Eq. 6:

$$WL (\%) = \frac{m_2 - m_3}{m_2} \times 100 \quad (6)$$



Figure 5: Screening termites test devices, with adjustments to EN 117 standard guidelines, for beech and pine wood controls and date palm woods. ©Kévin Candelier

2.7 Mineral composition analyses by ICP-OES

The aim of such an analysis is to determine the principal metals compounds (Ca, K, Mg, Si, etc.) contained in salt water treated and untreated palm wood samples. These comparisons between the both samples could allow us to better understand the chemical process reaction occurred in the treated palm wood structure during the salt water immersion. The results should allow obtaining a great data base to evaluate the improvement of salt water treated palm wood properties.

500 mg of dried powder sample was measured in platinum dish and placed into an oven. The oven was closed, the temperature was increased up to 250°C and then the oven was slightly open in order to evacuate the fumes. After that, the oven was closed and temperature will reached 500°C and maintained during 1.5 hours. The dish containing the sample was removed from the oven and the ash coming from the heated sample was humidify with few drops of pure water and reacted for 10 minutes with 2 ml of HNO₃ solution (50:50). The solution was filtered with lukewarm water, the filtrate was kept and the filter containing the residue was calcined in the oven at 500°C during 30 minutes. After, 2 ml of hydrofluoric acid solution (40:60) was added to the residue following by an evaporation performed and at 100°C. The sample was then calcined again into the oven at 500 °C during 10 minutes and then 1 ml of HNO₃ solution (50:50) was added to it. Finally the sample was filtered again with pure water to obtain a total substrate of 50 ml and its mineral composition was determined by ICP-OES Analysis.

3. RESULTS AND DISCUSSION

3.1 Chemical composition

Chemical composition was studied and showed that *Phoenix dactylifera* L. wood species have amounts of holocellulose, lignin and cellulose similar to those encountered in softwood and hardwood (Amirou et al. 2014). The contents of the chemical constituent of studied palm woods are detailed in Figure 6. It results that hemicellulose, cellulose, lignin and extractive contents of the both studied native *Phoenix dactylifera* L. species are ranged from 34.78 to 45.79 %, from 34.2 to 43.31%, from 13.14 to 18.15%, and from 4.93 to 6.35 %, respectively.

These results are consistent with those from other studies conducted on the chemical composition of Morocco, Egyptian and Tunisian date palm woods which found holocellulose contents ranged from 12.0 to 74.8 %, lignin contents ranged from 14.0 to 27.2 % and extractive contents comprised between 4.0 and 12.8 % (El Morsy, 1980; Khristova et al., 2005; Bendahou et al., 2007; Khiari et al., 2010).

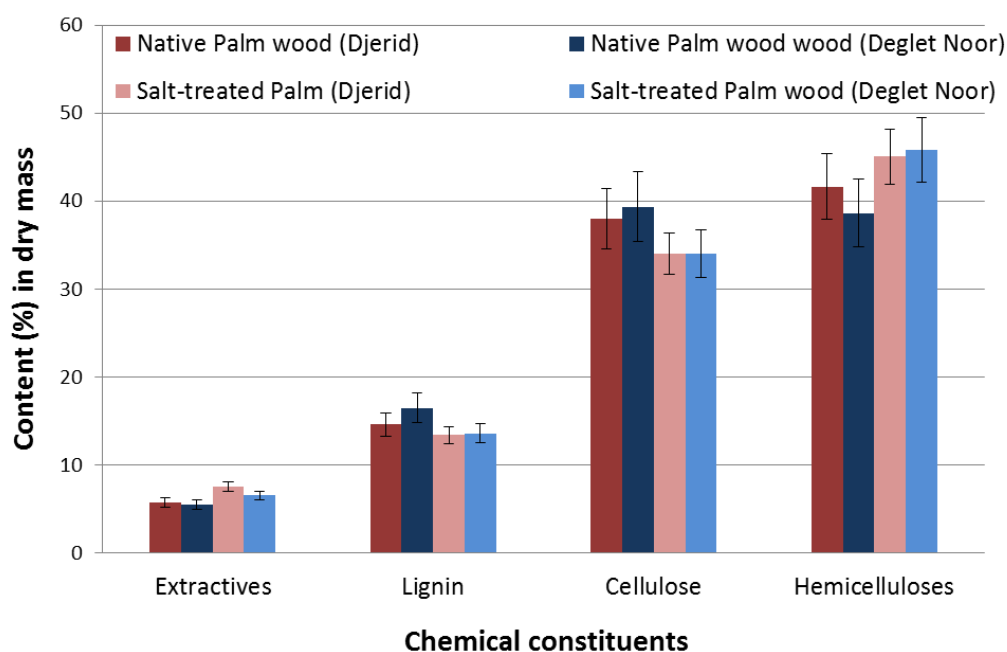


Figure 6: Contents of the different chemical constituents of *Kentichi* and *Deglet Noor* varieties palm woods.

According to [Figure 6](#), lignin and cellulose contents are more abundant in the control samples, whereas hemicelluloses and extractive contents are more abundant in the natural salt-treated palm wood samples and that for the both palm wood varieties (*Kentichi* and *Deglet Noor*). However, these tendencies are not really statistically valuable. This chemical composition modification between native and treated samples could be explained by the chemical reaction caused by several minerals contained in the salt water as chlorine (Cl), sodium (Na) and sulfur (S) during wood exposure to such a water [Guellela et al. \(2010\)](#).

3.2 Physical properties

Air-dried Density (D_{m12}) is commonly used to compare different woods. Basic Density (D_b), water saturated Density (D_s) and Air-dried Density (D_{m12}) have been measured on each palm wood sample. Average values of these different densities are presented in [Table 1](#).

The results obtained through this study are in agreement with those from [Almi et al. \(2015\)](#) who found that native Algerian date palm wood has a Basic Density ranged from 160 to 635 kg/m³ and a Water-saturated Density ranged from 838 to 995 kg/m³, according to the position of the wood in the tree (trunk, rachis, petiole, etc.).

[Table 1](#) shows the significant increase of the air dried and water-saturated densities which increase for the both palm wood species after the wood salt water immersion. This densification of salt-treated palm woods could be due to the impregnation of some mineral compounds into the wood during the wood exposition in the Lake Chot Djerid.

Table 1: Average values of Basic, Water-saturated and Air-dried densities of *Kentichi* and *Deglet Noor* palm woods.

Palm wood samples	Densities (%)					
	Basic Density (D_b)	SD^*	Water-saturated Density (D_s)	SD^*	Air-dried Density (D_{m12})	SD^*
Native Palm wood (<i>Djerid</i>)	0,210	0,13	0,829	0,21	0,217	0,12
Native Palm wood wood (<i>Deglet Noor</i>)	0,207	0,04	0,889	0,31	0,215	0,08
Salt-treated Palm (<i>Djerid</i>)	0,227	0,46	0,913	1,12	0,440	0,08
Salt-treated Palm wood (<i>Deglet Noor</i>)	0,243	0,02	1,057	0,35	0,361	0,02

* Standard Deviation.

3.3 Mechanical properties

The mechanical test results of each palm wood sample issued conditioned at a temperature of 20°C and 65% RH are shown in [Table 2](#). Modulus of rupture (MOR) reflects the maximum capacity in bending or in compression and is proportional to the maximum moment borne by a sample. Results of flexural strength showed that modulus of rupture (MOR) was ranged from 11.1 to 15.8 MPa for Native Palm wood and that it was improved to 20.7-30.2 MPa by natural saltwater treatment. Similar trend was observed concerning the flexural strength highlighting compression strength (MOR) improvement from 3.3 - 5.9 MPa to 10.3 - 14.3 MPa after saltwater immersion ([Table 2](#)). No results about natural saltwater treated palm woods are available in the literature, but the mechanical properties obtained on the native *Phoenix dactylifera* L. woods through this study are consistent with those issued from previous works. [Amirou et al. \(2014\)](#) and [Keramat Jahromi et al. \(2008\)](#) found date palm wood compression strength of 2.338 MPa

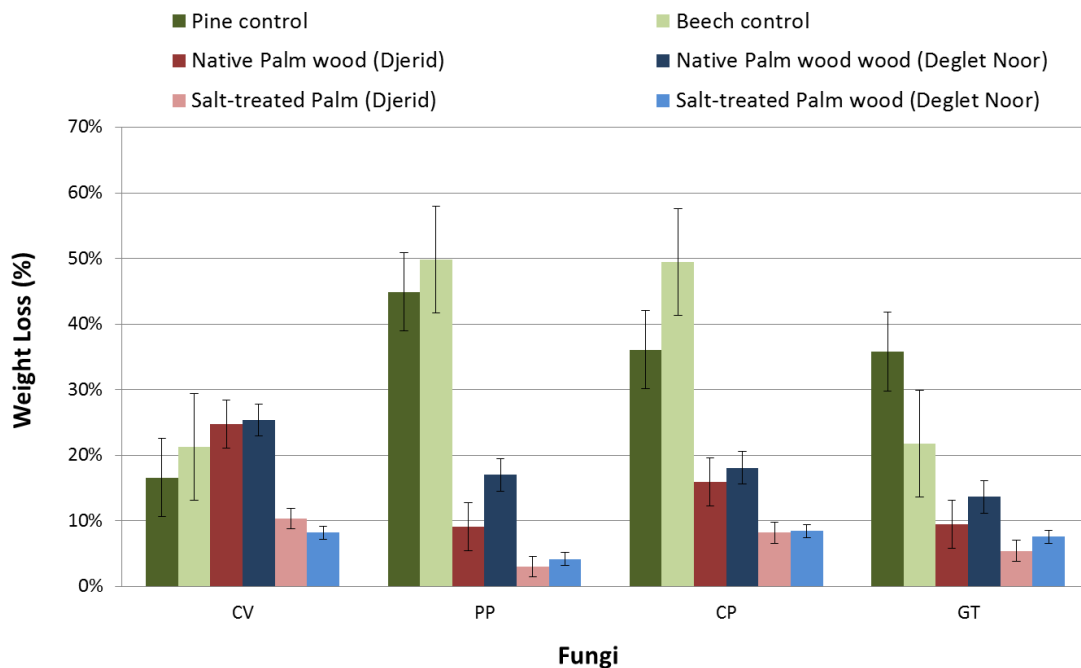
and 4.458 MPa, respectively. In addition, [Keramat Jahromi et al. \(2008\)](#) found that flexural strength of date palm trunk was around 7.166 MPa.

Table 2: Average values of Modulus of Rupture (MOR) in bending and compression strength of *Kentichi* and *Deglet Noor* palm woods.

Palm wood samples	Bending strength (MOR)		Compression strength (MOR)	
	Average values (MPa)	SD	Average values (MPa)	SD
Native Palm wood (<i>Djerid</i>)	11,1	2,9	3,3	1,04
Native Palm wood wood (<i>Deglet Noor</i>)	15,8	4,13	5,9	1,17
Salt-treated Palm (<i>Djerid</i>)	20,7	4,81	10,3	3,01
Salt-treated Palm wood (<i>Deglet Noor</i>)	30,2	6,26	14,3	1,56

3.3 Decay resistance

The effect of natural salt- water treatment on *Djerid* and *Deglet Noor* Tunisian date palm wood species was studied on the wood durability towards both fungi and termites. The tests were performed without accelerated ageing according to the methods described in the [EN 350](#) and the results are exposed in [Figure 7](#). The median mass loss on the two native Palm wood specimens was 25.37 % for *Coriolus versicolor*, 18.11 % for *Coniophora puteana*, 17.03 % for *Poria placenta* and 13.69 % for *Gloeophyllum trabeum*. The Mass Losses of the controls allow the test validation. Similar results have been found by previous study showing that white-rot fungi caused significantly more weight loss than brown-rot fungi on *Phoenix dactylifera* L. ([Adaskaveg et al., 1991](#)).



CV: *Coriolus versicolor*; PP: *Poria placenta*; CP: *Coniophora puteana*; GT: *Gloeophyllum trabeum*.

Figure 7: Average WL of native and natural saltwater-treated *Kentichi* and *Deglet Noor* varieties palm woods.

It appears clearly that natural salt-water treatment of *Palm* wood allows to decrease the weight loss due to fungal degradation. The median mass loss on the two modified *Palm* wood specimens was 9.37 % for *Coriolus versicolor*, 8.49 % for *Coniophora puteana*, 4.17 % for *Poria placenta* and 7.58 % for *Gloeophyllum trabeum*. The Mass Losses of the controls allow the test validation. According to EN 350 (2016), *Djerid* and *Deglet Noor* Tunisian date palm wood could be classified as moderately durable ($10 < WL \leq 15$) against *Gloeophyllum trabeum* and slightly durable ($15 < WL \leq 30$) towards *Coriolus versicolor*, *Coniophora puteana*, and *Poria placenta*. Natural saltwater treatment improves the decay resistance of the two tested palm wood classifying it as durable ($5 < WL \leq 10$) towards *Coriolus versicolor*, *Coniophora puteana* and *Gloeophyllum trabeum*, and as very durable ($WL \leq 5$) against *Poria placenta*.

Considering the little difference between the mean and median mass losses for each test fungus, durability against brown rot fungi does not appear to be variable. The improvement of decay resistance of modified palm woods is probably due to their higher extractive contents than those of native wood and their impregnation of some mineral compounds coming from their salt-water expositions (Gomez et al., 2016).

Treated palm wood samples had markedly lower weight losses than did the respective native blocks, although visual examination showed that fungal growth and damage were considerably reduced on the treated blocks. This added durability might be useful for preventing the development of decay in service, although it is difficult to assess the benefits of such salt loadings when the risk of leaching is high. Applying an additional oil-based treatment to the wood might retard leaching loss and maintain the protection (Schneider et al. 1997).

3.4 Termite resistance

The termite resistance test is valid as the *Pinus sylvestris* controls present a strong attack (ML = 21.96 %, visual rating 4) and above 50% of the termite workers remain alive at the end of the test. After 4 weeks of native *Djerid* and *Deglet Noor* Tunisian date palm woods exposure to termites, the mean survival rate of the insects was 20.00% and 21.33% respectively, and termites had got through all samples causing tunneling and heavy degradations (ML = 41.40 % and ML = 46.52 %, respectively). Native Tunisian date palm woods are classified as sensible towards termites. However that salt water treatment allows to obtain lower mass loss due to termites degradation (ML = 30.57 % for *Djerid* and ML = 32.60 % for *Deglet Noor*) and a higher termite mortality (6.00 % for *Djerid* and 8.67 % for *Deglet Noor*), modified *Djerid* and *Deglet Noor* Tunisian date palm woods are also classified as sensible towards termites. All results are detailed in Table 3.

Table 3: Mean values of Mass Loss (%) due to termites degradation, termites survival rates (%) and visual rating according to EN 117 (2013).

Palm wood samples	Mass Loss (%)	SD (%)	Survival rates (%)	SD (%)	Visual Rating
					According to EN 117
Native Palm wood (<i>Djerid</i>)	40.41	0.06	20.00	4.00	4
Native Palm wood wood (<i>Deglet Noor</i>)	46.52	0.15	21.33	3.06	4
Salt-treated Palm (<i>Djerid</i>)	30.57	0.09	6.00	5.29	4
Salt-treated Palm wood (<i>Deglet Noor</i>)	32.60	0.12	8.67	1.53	4
Beech control	26.60	0.02	63.33	9.45	4
Pine control	21.96	0.04	70.00	11.14	4

3.5 Mineral compositions and impact on palm wood durability

The mineral composition analyses conducted on native and salt-treated palm wood samples by ICP-OES are currently underway. This part is based only on results from the bibliography. However, it allows us to have some ways to explain the improvement of the properties of palm woods treated by salt water immersion process.

3.5.1 Mineral composition of Native Date palm wood

According to literature, Date palm trunk wood is mainly composed by carbon (75.86%) and oxygen (20.45%) (Khiari *et al.*, 2010). Date palm wood raw materials has also an important quantity of ashes (Khiari *et al.*, 2010), mainly constituted by Ca, Cl, K, S, and Na. In addition, the absolute amount of silicon is very low (Khristova *et al.*, 2005). This composition can be attributed to the chemical composition of the soil environment in which the plants are growing (Khiari *et al.*, 2010). Other minor constituents in trace are also present in native date palm wood included Mg and Si (Agoudjil *et al.*, 2011).

3.5.2 Mineral composition of Water from the Lake Chot Djerid

A previous study conducted by Guellela *et al.* (2010) shown that the water composing the Lake Chot Djerid around Tozeur area contained between 2 and 3 g/l of salt components, mainly composed by: SO_4 ($\approx 53\%$), Na ($\approx 20\%$), Cl ($\approx 12\%$), Ca ($\approx 10\%$), HCO_3 ($\approx 3\%$), K ($\approx 1\%$) and Mg ($\approx 1\%$).

3.5.3 Mineral composition of Salt-Treated palm wood and its influence on impact on wood decay resistance.

According to the previous typical salt water composition, Schneider *et al.* (1997) showed, by Energy-dispersive X-ray spectroscopy (EDXA), that Douglas fir wood immersed into such a water have a higher content of chlorine (Cl), sodium (Na), potassium (K), and sulfur (S) than the respective native wood. Scanning Electron Microscope (SEM) examination highlighted also the present and crystal deposits into the wood of aluminum (Al), calcium (Ca), chlorine, magnesium (Mg) and silica (Si) (Schneider *et al.*, 1997). Wood exposed to salt water appeared to contain high levels of sodium chloride (NaOH) and also a small amount of potassium chloride (KCl). The pH of these last compounds are around 11-12 and 5.5, respectively. The high content of NaOH present into the salt treated wood samples could increase the pH of wood and then inhibit the mycelium growth on the treated samples, knowing that the optimal pH for wood decay mycelium growth is comprised between 5 and 6 (Bozkurt *et al.*, 1993). This statement could explain the fact that salt-treated date palm wood is more resistant against Brown Rot [*Coniophora puteana* (CP), *Poria placenta* (PP) and *Gloeophyllum trabeum* (GT)] than White Rot [*Coriolus versicolor* (CV)]. Indeed, Highley (1976) found the cellulose and hemicellulases of brown-rot fungi to be more tolerant of low pH and more sensitive to high pH than those of white-rot fungi. Medium acidity was shown to exert a strong influence on the rate of lignin degradation by white-rot fungus, little degradation occurred below pH 3 or above pH 5 (Kirk *et al.*, 1978). While decaying wood, brown rot fungi create a low pH, primarily by oxalic acid production; however, unlike white-rot fungi, which also produce oxalic acid, the brown-rots are unable to metabolize this product (Takao, 1965). In addition, Schneider *et al.* (1997) noted that salt leaching from the salt-treated wood blocks during the test killed fungi on the feeder strips. Their EDXA examination showed elevated levels of sulfur and a strong odor of sulfur was emitted from fresh samples, the source of this element could not be determined. Sulfur appeared to be resistant to leaching (Schneider *et al.* 1997). This salt leaching shows that the permanency of soluble salts is only satisfactory in the indoor where the material is not exposed to water. As expected, Catelli *et al.* (2016) proved that the salts were washed out by precipitation from the outdoor areas. This added durability might be useful for preventing the development of decay in service, although it is difficult to assess the benefits of such salt loadings when the risk

of leaching is high. Applying an additional oil-based treatment to the wood might retard leaching loss and maintain the protection (Schneider *et al.* 1997).

3.5.4 Mineral composition of Salt-Treated palm wood and its influence on impact on wood termite resistance.

Termite resistance of salt-treated palm wood compared to those of native samples is less obvious than its decay resistance improvement. However, the termite mortality is higher for treated than those of native wood sample. This result can be explained by the followings:

(i) The high proportion of Silica contained into the treated palm wood influence termite damage and plays a significant role in its termite resistance (Dhawan *et al.*, 2007).

(ii) High ash content reduces the food value of bamboo for termites because it is not absorbed in the gut body and passes through (Hapukotuwa and Grace, 2011).

(iii) Some minerals in bamboo can have a toxic effect on the pests or disturb their physiology (Doi *et al.*, 1998; Kneip *et al.*, 2007). Termite damage was in relation to the chemicals of the wood samples and Subekti *et al.* (2015) showed that NaOH content present into wood sample have a high impact on termite mortality.

4. CONCLUSION

Native Tunisian *Djerid* and *Deglet Noor* date palm woods showed a low durability: moderately durable against *Gloeophyllum trabeum* and slightly durable towards *Coriolus versicolor*, *Coniophora puteana*, and *Poria placenta* attacks, and were classified “sensible” against termites. Results showed that natural saltwater treatment in the Lake Chot Cherid of both date palm woods allowed to improve the wood decay resistance until the classes “durable” towards *Coriolus versicolor*, *Coniophora puteana* and *Gloeophyllum trabeum*, and “very durable” against *Poria placenta*. However, this ancestral preservation process did not allow to improve significantly the date palm wood resistance against termites. In addition, results showed a significant increase of the air-dried density of palm wood samples which increases after the wood salt water immersion. Salt water treatment allows improving greatly the palm wood MOR in bending and paralleling to the fibers.

In conclusion, all qualification testing showed that the saltwater treatment performed on Tunisian date palm woods (*Djerid* and *Deglet Noor* species) conferred to the wood material a good technological value. However, this process is still largely unknown and needs to be more controlled and optimized. In addition, the added durability might be useful for preventing the development of decay in service, although it is difficult to assess the benefits of such salt loadings when the risk of leaching is high. Applying an additional oil-based treatment to the wood might retard leaching loss and maintain the protection. In this sense, it could be interesting now to think about a new industrial process based on this natural and ancestral method to valorize Tunisian date palm sector and commercialized new date palm wood materials.

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6. REFERENCES

- Abu-Sharkh B F, Hamid H (2004): Degradation study of date palm fibre/polypropylene composites in natural and artificial weathering: Mechanical and thermal analysis. *Polym. Degrad. Stab.*, **85**, 967-973.
- Adaskaveg J E, Blanchette R A, Gilbertson R L (1991): Decay of date palm wood by white-rot and brown-rot fungi. *Canadian Journal of Botany*, **69** (3), 615-629.
- Agoudjil B, Benchabane A, Boudenne A, Ibos L, Fois M (2011): Renewable materials to reduce building heat loss: Characterization of date palm wood. *Energy and Buildings*, **43**, 491-497.
- Almi K, Lakel S, Benchabane A, Kriker A (2015): Characterization of Date Palm Wood Used as Composites Reinforcement. *Acta Physica Polonica A*, **127** (4), 1072-1074.
- Amirou S, Zerizer A, Pizzi A (2014): Investigation of Chemical, Physical and Mechanical Properties of Algerian Date Palm Wood. *Materials Testing*, **56** (3), 236-240.
- Babiker M E, Aziz A R, Heikal M, Yusup S, Abakar M (2013): Pyrolysis characteristics of *Phoenix dactylifera* date palm seeds using thermogravimetric analysis (TGA). *Int. J. Environ. Sci. Dev.*, **4**, 521-524.
- Bendahou A, Dufresne A, Kaddami H., Habibi Y (2007): Isolation and structural characterization of hemicelluloses from palm of *Phoenix dactylifera* L. *Carbohydrate Polymers*, **68**, 601-608.
- Bozkurt AY, Göker Y, Erdin N (1993): *Emprenye tekniği*. Istanbul Üniversitesi Orman Fakültesi, Yayın No: 3779, Istanbul.
- Catelli E, Bănică F G, Bănică A (2016): Salt efflorescence in historic wooden buildings. *Heritage Science*, **4**, 31.
- Dhawan S, Mishra S C, Dhawan S (2007): A Study of Termite Damage in Relation to Chemical Composition of Bamboos. *Indian For.*, **133** (3), 411-419.
- Doi S, Takahashi T, Yoshimura T, Kubota M, Adachi A (1998): Attraction of steamed Japanese Larch [*Larix leptolepis* (Sieb.et Zucc.) Gord] heartwood to the subterranean termite *Coptotermes formosanus Shiraki* (Isoptera: Rhinotermitidae). *Holzforschung*, **52**, 7-12.
- Eissa A H A (2013): Quality characteristics for agricultural residues to produce briquettes. In *Proceedings of the International Conference on Agricultural, Biotechnology, Biological and Biosystems Engineering*, New York, NY, USA, 5-6 June 2013.
- El Morsy M M S (1980): Studies on the rachis of Egyptian date palm leaves for hardboard production. *Fibre Science and Technology*, **13** (4), 317-321.
- El-Mously H I (1997): The rediscovery of local raw materials: New opportunities for developing countries. *Ind. Environ.*, **20**, 17-20.
- Ellwoode L and Ecklund B A (1959): Bacterial attack of pine logs in pond storage. *Forest Prod. J.* **9**, 283-292.

EN 117 (2013): Wood preservatives - Determination of toxic values against *Reticulitermes* species (European termites) (laboratory method). European committee for standardization.

EN 350 (2016): Durability of wood and wood-based products — Testing and classification of the durability to biological agents of wood and wood-based materials. European committee for standardization.

EN 408 (2010): Standard - Timber structures - structural timber and glued laminated timber - determination of some physical and mechanical properties (+A1:2012). European committee for standardization.

Gómez R, Asencio A D, Picón J M, Del Campo R, Arce M I, Del Mar Sánchez-Montoya M, Suárez M L, Vidal-Abarca M R (2016): The effect of water salinity on wood breakdown in semiarid Mediterranean streams. *Sci Total Environ.*, **541**, 491-501.

Guellala R, Moumni L, Hédi Inoubli M (2010): Characteristics of the Lower Cretaceous aquifer of the Djerid, Tunisia [In French]. *Geo-Eco-Trop*, **34**, 87 – 92.

Hegazy S S, Ahmed K (2015): Effect of date palm cultivar, particle size, panel density and hot water extraction on particleboards manufactured from date palm fronds. *Agriculture* 2015, **5**, 267-285.

Hapukotuwa N K and Grace J K (2011): Comparative study of the resistance of six Hawaii-grown bamboo species to attack by the subterranean termites *Coptotermes formosanus* Shiraki and *Coptotermes gestroi* (Wasmann) (Blattodea: Rhinotermitidae). *Insects*, **2**, 475-485.

Highley TL (1976): Hemicellulases of white-rot and brown-rot fungi in relation to host preferences. *Matt. Org.*, **11**(1), 33-46.

ISO 13061-2 (2014): Physical and mechanical properties of wood - Test methods for small clear specimens - Part 2: determination of density for physical and mechanical tests, 5 p. International Organization for Standardization.

ISO 4471 (1982): Wood - Sampling sample trees and logs for determination of physical and mechanical properties of wood in homogeneous stands. International Organization for Standardization

Johnson B R, Ibach R E, Baker AJ (1992): Effect of salt water evaporation on tracheid separation from wood surfaces. *Forest Prod. J.* **42**(7/8), 57-59.

Keramat Jahromi M., Rafiee S., Jafari A., Ghasemi Bousejin M.R., Mirasheh R., Mohtasebi S.S. (2008): Some physical properties of date fruit (cv. Dairi). *Int. Agrophys.*, **22**, 221-224

Khiari R, Mhenni M F, Belgacem M N, Mauret E (2010): Chemical composition and pulping of date palm rachis and *Posidonia oceanica* – A comparison with other wood and non-wood fibre sources. *Bioresource Technology*, **101**, 775-780.

Khiari R, Mauret E, Belgacem M N, Mhemmi F (2011) : Tunisian date palm rachis used as an alternative source of fibres for papermaking applications. *BioResource*, **6**, 265–281.

- Khristova P, Kordsachia O, Khider T (2005): Alkaline pulping with additives of date palm rachis and leaves from Sudan. *Bioresource Technology*, **96**, 79–85.
- Kirk T K, Shultz E, Connors W J, Lorenz L F, Zeikus J G (1978): Influence of cultural parameters on lignin metabolism by *Phanerochaete chrysosporium*. *Arch. Microbiol.*, **117**, 277-285.
- Kneip C, Lockhart P, Voss C, Maier U G (2007): Nitrogen fixation in eukaryotes—new models for symbiosis. *BMC Evol. Biol.*, **7**: 55.
- Nasser RA and Al-Mefarrej H A (2011): Midribs of date palm as a raw material for wood-cement composite industry in Saudi Arabia. *World Appl. Sci. J.*, **15**(12), 1651-1658.
- Nasser R A (2014): An evaluation of the use of midribs from common date palm cultivars grown in Saudi Arabia for energy production. *BioResource*, **9**, 4343-4357.
- Nasser R A, Hiziroglu H, Abdel-Aal M A, Al-Mefarrej H A, Shetta N D, Aref I M (2015): Measurement of some properties of pulp and paper made from date palm midribs and wheat straw by soda-AQ pulping process. *Measurement*, **62**, 179-186.
- Nasser R A, Salem M Z M, Hiziroglu S, Al-Mefarrej H A, Mohareb A S, Alam M, Aref I M (2016): Chemical Analysis of Different Parts of Date Palm (*Phoenix dactylifera* L.) Using Ultimate, Proximate and Thermo-Gravimetric Techniques for Energy Production. *Energies*, **9**, 374, 1-14.
- Rhouma A (1993) : Le palmier dattier en Tunisie : Un secteur en pleine expansion. In: Ferry M. (ed.), Greiner D. (ed.). *Le palmier dattier dans l'agriculture d'oasis des pays méditerranéens*. Zaragoza : CIHEAM, 1993.p.85-104 (Options Méditerranéennes : Série A. Séminaires Méditerranéens; n. 28).
- Schneider P F, Freitag C M, Morrell J J (1997): Decay resistance of saltwater exposed Douglas-fir piles. *Wood and Fiber Science*, **19**(4), 370-374.
- Subekti N, Yoshimura T, Rokhman F, Mastur Z (2015): Potential for subterranean termite attack against five bamboo species in correlation with chemical components. *Procedia Environmental Sciences*, **28**, 783-788.
- Takao S (1965): Organic acid by basidiomycetes I. Screening of acid-producing strains. *Appl. Microbiol.*, **13**, 732-737.
- Usman A R, Abduljabbar A, Vithanag M, Ok Y S, Ahmad M, Ahmad M, Elfaki J, Abdulazeem S S, Al-Wabel M I (2016): Biochar production from date palm waste: Charring temperature induced changes in composition and surface chemistry. *J. Anal. Appl. Pyrolysis*, **115**, 392-400.
- Zabel R A and Morrell J.J (1992): *Wood microbiology: Decay and its prevention*. Academic Press, San Diego, CA. 478 pp.